

Optimization

Class Note 1.

NIT Manipur

Introduction

Many real-life problems require us to make the best decision from several available choices. For example,

1. A company wants to earn the maximum profit.
2. A factory wants to minimize the production cost.
3. A farmer wants to obtain the maximum crop yield.

The process of finding the best solution is called **optimization**.

What is Optimization?

Optimization is the process of finding the best possible solution to a problem by maximizing or minimizing an objective function while satisfying all given constraints.

Example 1: A shopkeeper sells two products.

1. Profit from Product A = Rs. 50 per unit
2. Profit from Product B = Rs. 40 per unit

The shopkeeper wants to produce products in such a way that the total profit is maximum. This is an optimization problem.

Optimization problems can be classified according to the nature of the objective function, the constraints, and the decision variables.

Classification of Optimization Problems

1. **Linear Optimization (Linear Programming Problem):** A linear optimization problem is one in which both the objective function and all constraints are linear.

Example: Maximize $Z = 3x + 2y$

Subject to $x + y \leq 5$, $2x + y \leq 8$, $x, y \geq 0$.

2. **Nonlinear Optimization (Non linear Programming Problem):** A nonlinear optimization problem has a nonlinear objective function or at least one nonlinear constraint.

Example: Minimize $f(x, y) = x^2 + y^2$. Subject to $x + y \geq 2$.

3. Constrained Optimization

In constrained optimization, the objective function is optimized while satisfying one or more constraints.

General Form:

$$\begin{aligned} &\text{Minimize or Maximize } f(x) \\ &\text{Subject to } g_i(x) \leq 0, \quad i = 1, 2, \dots, m. \end{aligned}$$

4. Unconstrained Optimization

In unconstrained optimization, there are no constraints on the decision variables.

Example: Minimize $f(x) = x^2 - 4x + 5$.

The objective is simply to find the minimum or maximum value of the function.

5. **Convex Optimization:** A convex optimization problem has a convex objective function and a convex feasible region. An important property is:

Every local minimum is also the global minimum.

Convex optimization problems are generally easier to solve than non-convex problems.

6. **Integer Optimization:** In integer optimization, some or all decision variables are restricted to integer values.

Example: The number of buses, workers, or machines must be whole numbers.

7. **Dynamic Optimization (Dynamic Programming)**

Dynamic optimization solves a problem by dividing it into several stages.

Each decision made at one stage affects the decisions in later stages.

Many optimization problems involve limited resources, such as labour, raw materials, money, or machine time. When both the objective function and the restrictions are linear, we use a mathematical technique called Linear Programming to find the best solution.

What is Linear Programming (LP)?

Definition: Linear Programming (LP) is a mathematical technique used to find the maximum or minimum value of a linear function while satisfying a set of linear constraints.

Applications of Linear Programming: Linear Programming is used in Business, Manufacturing, Transportatio, Agricultur, Banking, Production Planning and Scheduling etc.

Linear Programming Problem (LPP)

Definition: A **Linear Programming Problem (LPP)** is a problem in which we maximize or minimize a linear objective function, and subject to a set of linear constraints.

Mathematical Description of a General Linear Programming Problem

A general Linear Programming Problem (LPP) can be stated as follows.

Find the values of the decision variables

$$x_1, x_2, \dots, x_n,$$

which maximize or minimize the following linear function:

$$Z = c_1x_1 + c_2x_2 + \dots + c_nx_n. \quad (1)$$

subject to the linear constraints

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &\leq (=, \geq) b_1, \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &\leq (=, \geq) b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &\leq (=, \geq) b_m. \end{aligned} \right\} \quad (2)$$

together with the non-negative restrictions

$$x_1, x_2, \dots, x_n \geq 0. \quad (3)$$

where a_{ij} , b_i , c_j are known constants and $x_1, x_2, \dots, x_n$ are the decision variables.

The function Z given in (1) is called the **objective function**, the equations or inequalities given in (2) are called the **constraints** and the conditions given in (3) are called the **non-negative restrictions**.

Matrix Form of a Linear Programming Problem

The above Linear Programming Problem can also be written in matrix form as

$$\text{Optimize } Z = CX$$

subject to $AX \leq (=, \geq)b$, and $X \geq 0$, where $A = [a_{ij}]_{m \times n}$ is called the **coefficient**

matrix, $C = [c_1 \ c_2 \ \dots \ c_n]$ is called the **cost (or price) vector**, $b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$ is called

the **requirement (or resource) vector**, and $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ is called the **decision variable** **vector**.

Example 1: A shopkeeper sells two products, Product A and Product B. The profit earned from selling one unit of Product A is Rs. 50, and the profit earned from selling one unit of Product B is Rs. 40. To produce one unit of Product A, 2 hours of labour and 1 kg of raw material are required. To produce one unit of Product B, 1 hour of labour and 2 kg of raw material are required. The shopkeeper has only 100 hours of labour and 80 kg of raw material available. The problem is to determine how many units of Product A and Product B should be produced so that the total profit is maximum without exceeding the available labour and raw material.

Formulation of the Mathematical Model

Let x = number of units of Product A, and y = number of units of Product B.

Form the Objective Function

The profit from one unit of Product A is Rs. 50 and the profit from one unit of Product B is Rs. 40.

Therefore,

1. Profit from x units of Product A = $50x$
2. Profit from y units of Product B = $40y$

Hence, the total profit is

$$Z = 50x + 40y.$$

Since the objective is to obtain the maximum profit,

Maximize $Z = 50x + 40y.$

Form the Constraints: Labour Constraint

Each unit of Product A requires 2 hours of labour, so x units require $2x$ hours.

Each unit of Product B requires 1 hour of labour, so y units require y hours.

Therefore, the total labour required is $2x + y$.

Since only 100 hours of labour are available, $2x + y \leq 100$.

Raw Material Constraint

Each unit of Product A requires 1 kg of raw material, so x units require x kg.

Each unit of Product B requires 2 kg of raw material, so y units require $2y$ kg.

Therefore, the total raw material required is $x + 2y$.

Since only 80 kg of raw material is available, $x + 2y \leq 80$.

Step 3: Non-negative Restrictions

The number of products cannot be negative. Therefore, $x \geq 0$, $y \geq 0$.

Hence, the Linear Programming Problem is

Maximize	$Z = 50x + 40y$
subject to	$2x + y \leq 100,$
	$x + 2y \leq 80,$
	$x \geq 0, y \geq 0.$

Example 2 (Diet Problem): The objective of a diet problem is to determine the quantities of certain food items that should be consumed so that the required nutritional

needs are satisfied at the minimum possible cost. Suppose the diet consists of three food items: milk, green vegetables, and eggs. The nutrients considered are Vitamins A , B , and C . The amount of each vitamin contained in one unit of each food item, the minimum daily vitamin requirement, and the cost of each food item are given in the following table.

Vitamin	Liter of Milk	Kg of Vegetables	Dozen of Eggs	Minimum Daily Requi
A	1	1	10	1 mg
B	100	10	10	50 mg
C	10	100	10	10 mg
Cost (Rs./unit)	20	10	8	

Formulate a linear programming problem to determine the daily diet of minimum cost.

Solution:

Let x_1 = litres of milk consumed per day, x_2 = kilograms of vegetables consumed per day, x_3 = dozens of eggs consumed per day.

These are the **decision variables** of the problem.

The cost of one litre of milk is Rs. 20, one kilogram of vegetables costs Rs. 10, and one dozen eggs costs Rs. 8. Therefore, the total daily cost of the diet is $Z = 20x_1 + 10x_2 + 8x_3$.

Since our objective is to obtain the required nutrients at the *minimum possible cost*, the objective function is

$$\boxed{\text{Minimize } Z = 20x_1 + 10x_2 + 8x_3.}$$

Next, we formulate the constraints corresponding to the minimum daily requirement of each vitamin.

Vitamin A Constraint

Each litre of milk provides 1 mg of Vitamin A , each kilogram of vegetables provides 1 mg, and each dozen eggs provides 10 mg.

Hence, the total amount of Vitamin A obtained from the daily diet is

$$x_1 + x_2 + 10x_3 \text{ mg.}$$

Since at least 1 mg of Vitamin A is required every day, $x_1 + x_2 + 10x_3 \geq 1$.

Vitamin B Constraint

Each litre of milk contains 100 mg of Vitamin B , each kilogram of vegetables contains 10 mg, and each dozen eggs contains 10 mg.

Therefore, the total amount of Vitamin B obtained is $100x_1 + 10x_2 + 10x_3$ mg.

Since the minimum daily requirement is 50 mg, $100x_1 + 10x_2 + 10x_3 \geq 50$.

Vitamin C Constraint

Each litre of milk contains 10 mg of Vitamin C , each kilogram of vegetables contains 100 mg, and each dozen eggs contains 10 mg.

Hence, the total amount of Vitamin C obtained is $10x_1 + 100x_2 + 10x_3$ mg. Since at least 10 mg of Vitamin C is required daily,

$$10x_1 + 100x_2 + 10x_3 \geq 10.$$

Non-negativity Constraints

The quantities of food items consumed cannot be negative. Therefore,

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

Hence, the required linear programming problem is

$$\begin{aligned} \text{Minimize} \quad & Z = 20x_1 + 10x_2 + 8x_3, \\ \text{subject to} \quad & x_1 + x_2 + 10x_3 \geq 1, \\ & 100x_1 + 10x_2 + 10x_3 \geq 50, \\ & 10x_1 + 100x_2 + 10x_3 \geq 10, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Question 3: A resourceful home decorator manufactures two types of lamps, namely Lamp A and Lamp B. Both lamps are processed by two technicians: first a cutter and then a finisher. Lamp A requires 2 hours of the cutter's time and 1 hour of the finisher's time. Lamp B requires 1 hour of the cutter's time and 2 hours of the finisher's time. The cutter has 104 hours and the finisher has 76 hours available each month. The profit earned on one Lamp A is Rs. 6 and on one Lamp B is Rs. 11. Assuming that all the lamps produced can be sold, formulate the above problem as a Linear Programming Problem.

Solution: Formulation of the Mathematical Model

Let the decorator manufacture x_1 and x_2 lamps of types A and B, respectively.

The profit earned from one Lamp A is Rs. 6 and from one Lamp B is Rs. 11.

Therefore, the total profit (in Rs.) is $Z = 6x_1 + 11x_2$.

Since the objective is to maximize the profit, $\text{Maximize } Z = 6x_1 + 11x_2.$

The cutter requires 2 hours to prepare one Lamp A.

Therefore, the time required for preparing x_1 lamps of type A is $2x_1$ hours.

The cutter requires 1 hour to prepare one Lamp B.

Therefore, the time required for preparing x_2 lamps of type B is x_2 hours.

Hence, the total cutter's time required to prepare x_1 lamps of type A and x_2 lamps of type B is $2x_1 + x_2$ hours. Since the cutter has only 104 hours available,

$$2x_1 + x_2 \leq 104. \tag{4}$$

The finisher requires 1 hour to finish one Lamp A.

Therefore, the finishing time required for x_1 lamps of type A is x_1 hours.

The finisher requires 2 hours to finish one Lamp B.

Therefore, the finishing time required for x_2 lamps of type B is $2x_2$ hours.

Hence, the total finishing time required is $x_1 + 2x_2$ hours.

Since the finisher has only 76 hours available, $x_1 + 2x_2 \leq 76$.

Also, the number of lamps manufactured cannot be negative. Therefore,

$$x_1 \geq 0, \quad x_2 \geq 0. \tag{5}$$

Hence, the required Linear Programming Problem is

$$\begin{aligned} \text{Maximize} \quad & Z = 6x_1 + 11x_2 \\ \text{subject to} \quad & 2x_1 + x_2 \leq 104, \\ & x_1 + 2x_2 \leq 76, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

Thus, the above mathematical model represents the given Linear Programming Problem.

Question 5: A tyre factory produces three types of tyres, namely T_1 , T_2 , and T_3 . Three different types of chemicals, namely C_1 , C_2 , and C_3 , are required for their production. One tyre of type T_1 requires 2 units of chemical C_1 and 3 units of chemical C_3 . One tyre of type T_2 requires 3 units of chemical C_1 , 2 units of chemical C_2 , and 2 units of chemical C_3 . One tyre of type T_3 requires 5 units of chemical C_2 and 4 units of chemical C_3 . The factory has only 20 units of chemical C_1 , 25 units of chemical C_2 , and 30 units of chemical C_3 in stock. The profit obtained from selling one tyre of type T_1 is Rs. 6, one tyre of type T_2 is Rs. 10, and one tyre of type T_3 is Rs. 8. Assuming that the factory can sell all the tyres it produces, formulate the above problem as a Linear Programming Problem.

Solution. Formulation of the Mathematical Model

Let the factory produce x_1 , x_2 , and x_3 tyres of types T_1 , T_2 , and T_3 , respectively.

The given information can be represented in the following table.

Chemicals	T_1 (x_1)	T_2 (x_2)	T_3 (x_3)	Available
C_1	2	3	0	20
C_2	0	2	5	25
C_3	3	2	4	30
Profit (Rs./tyre)	6	10	8	—

The profit obtained from one tyre of type T_1 is Rs. 6, one tyre of type T_2 is Rs. 10, and one tyre of type T_3 is Rs. 8.

Therefore, the total profit (in Rs.) is

$$Z = 6x_1 + 10x_2 + 8x_3. \quad (6)$$

Since the objective is to maximize the profit,

Maximize $Z = 6x_1 + 10x_2 + 8x_3.$

Chemical C_1 Constraint

One tyre of type T_1 requires 2 units of chemical C_1 .

Therefore, x_1 tyres require $2x_1$ units of C_1 .

One tyre of type T_2 requires 3 units of chemical C_1 . Therefore, x_2 tyres require $3x_2$ units of C_1 . Tyre T_3 does not require chemical C_1 .

Hence, the total quantity of chemical C_1 required is $2x_1 + 3x_2$.

Since only 20 units of chemical C_1 are available,

$$2x_1 + 3x_2 \leq 20. \quad (7)$$

Chemical C_2 Constraint

One tyre of type T_2 requires 2 units of chemical C_2 . Therefore, x_2 tyres require $2x_2$ units of C_2 . One tyre of type T_3 requires 5 units of chemical C_2 . Therefore, x_3 tyres require $5x_3$ units of C_2 . Hence, the total quantity of chemical C_2 required is $2x_2 + 5x_3$. Since only 25 units of chemical C_2 are available, $2x_2 + 5x_3 \leq 25$.

Chemical C_3 Constraint

One tyre of type T_1 requires 3 units of chemical C_3 . Therefore, x_1 tyres require $3x_1$ units of C_3 . One tyre of type T_2 requires 2 units of chemical C_3 . Therefore, x_2 tyres require $2x_2$ units of C_3 . One tyre of type T_3 requires 4 units of chemical C_3 .

Therefore, x_3 tyres require $4x_3$ units of C_3 . Hence, the total quantity of chemical C_3 required is $3x_1 + 2x_2 + 4x_3$. Since only 30 units of chemical C_3 are available, $3x_1 + 2x_2 + 4x_3 \leq 30$. Since the number of tyres produced cannot be negative,

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0. \quad (8)$$

Hence, the required Linear Programming Problem is

$$\begin{aligned} \text{Maximize} \quad & Z = 6x_1 + 10x_2 + 8x_3 \\ \text{subject to} \quad & 2x_1 + 3x_2 \leq 20, \\ & 2x_2 + 5x_3 \leq 25, \\ & 3x_1 + 2x_2 + 4x_3 \leq 30, \\ & x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0. \end{aligned}$$

Thus, the above mathematical model represents the given Linear Programming Problem.

Some Important Definitions

1. **Solution of a Linear Programming Problem (LPP):** A set of values assigned to the decision variables $x_1, x_2, \dots, x_n$ that satisfies all the constraints of the Linear Programming Problem is called a **solution** of the LPP.
2. **Feasible Solution of a Linear Programming Problem (LPP):** A solution that satisfies all the constraints as well as the non-negative restrictions $x_1, x_2, \dots, x_n \geq 0$ is called a **feasible solution** of the LPP.
3. **Optimal (or Optimum) Solution:** A feasible solution that gives the best possible value of the objective function (maximum or minimum, as required) is called an **optimal (or optimum) solution** of the LPP.
4. **Unbounded Solution:** If the value of the objective function can be increased indefinitely (for a maximization problem) or decreased indefinitely (for a minimization problem) without violating any of the constraints, then the problem is said to have an **unbounded solution**.
5. **Fundamental Extreme Point Theorem:** If an optimal solution of a Linear Programming Problem exists, then at least one optimal solution occurs at an extreme point (or corner point) of the feasible region.

Basic Solution

Consider the system of linear equation $AX = b$, where A is an $m \times n$ coefficient matrix, X is the vector of unknowns, and b is the constant vector. Assume that $n > m$ and

$$\text{rank}(A) = \text{rank}(A|b) = m.$$

This implies that the system is consistent and all the equations are linearly independent.

Definition: A **basic solution** of the system $AX = b$ is obtained by selecting any m linearly independent columns of the coefficient matrix A . The variables corresponding to these columns are called the *basic variables*, while the remaining $n - m$ variables are set equal to zero and are called the *non-basic variables*. The values of the basic variables are then determined by solving the resulting system of m equations. Equivalently, if B denotes the non-singular matrix formed by the selected m columns of A , then the basic variables are obtained from $X_B = B^{-1}b$.

Thus, every basic solution contains at least $n - m$ variables equal to zero.

Example 6: Consider the system

$$\begin{aligned}x_1 + x_2 + x_3 &= 4, \\2x_1 + x_2 + 3x_3 &= 7.\end{aligned}$$

Here, $m = 2, n = 3$. Since there are three variables and only two equations, one variable must be chosen as a non-basic variable. Choose x_3 as the non-basic variable and set $x_3 = 0$. The system becomes

$$\begin{aligned}x_1 + x_2 &= 4, \\2x_1 + x_2 &= 7.\end{aligned}$$

Subtracting the first equation from the second, $x_1 = 3$. Substituting into the first equation, $x_2 = 1$. Hence, the basic solution is $(x_1, x_2, x_3) = (3, 1, 0)$.

In this solution, x_1 and x_2 are the **basic variables**, x_3 is the **non-basic variable**.

Similarly, we can choose a different variable as the non-basic variable.

For example, let $x_2 = 0$.

Then the system becomes

$$\begin{aligned}x_1 + x_3 &= 4, \\2x_1 + 3x_3 &= 7.\end{aligned}$$

Solving these equations, $x_3 = -1$, $x_1 = 5$. Hence, another basic solution is $(x_1, x_2, x_3) = (5, 0, -1)$.

Notice that a basic solution is not necessarily non-negative. A basic solution is called a **basic feasible solution** only if all its variables are non-negative.

1 Basic Feasible Solution

Definition: A **basic feasible solution (BFS)** is a solution of a linear programming problem that is both a *basic solution* and a *feasible solution*. In other words, a basic solution is called a basic feasible solution if it satisfies all the constraints of the linear programming problem, including the non-negativity restrictions. Equivalently, a feasible solution in which the vectors corresponding to the non-zero variables are linearly independent is called a basic feasible solution.

Since there are m constraints, at most m variables can be basic. Therefore, every basic feasible solution contains at most m positive variables and at least $n - m$ variables must be equal to zero, where n is the total number of variables.

Example 7: Consider the linear programming problem

$$\begin{aligned}x_1 + x_2 + x_3 &= 4, \\2x_1 + x_2 + x_4 &= 5, \\x_1, x_2, x_3, x_4 &\geq 0.\end{aligned}$$

Here, $m = 2, n = 4$. Hence, every basic feasible solution must have at least $n - m = 4 - 2 = 2$ variables equal to zero. Choose $x_3 = 0, \quad x_4 = 0$. Then the system becomes

$$\begin{aligned}x_1 + x_2 &= 4, \\2x_1 + x_2 &= 5.\end{aligned}$$

Solving, $x_1 = 1, x_2 = 3$. Therefore, $(x_1, x_2, x_3, x_4) = (1, 3, 0, 0)$.

Since all the variables are non-negative, this is a **basic feasible solution**.

Definition: A **basic feasible solution** is called an **optimal basic feasible solution** if it gives the optimum (maximum or minimum) value of the objective function among all feasible solutions. Thus, an optimal basic feasible solution is the required solution of the linear programming problem.

Definition: The variables corresponding to the selected linearly independent columns of the coefficient matrix are called the **basic variables**. These variables are obtained by solving the system of equations and may take positive values.

For example, in the basic feasible solution

$$(x_1, x_2, x_3, x_4) = (1, 3, 0, 0),$$

the variables x_1 and x_2 are the basic variables.

Definition: The variables that are set equal to zero while obtaining a basic solution are called the **non-basic variables**. In every basic feasible solution, there are at least $n - m$ non-basic variables.

In the above example, $x_3 = x_4 = 0$, hence x_3 and x_4 are the non-basic variables.

Definition: A basic feasible solution is said to be **non-degenerate** if all its basic variables are strictly positive. In other words, none of the basic variables is equal to zero. For example,

$$(x_1, x_2, x_3, x_4) = (1, 3, 0, 0)$$

is a non-degenerate basic feasible solution because both basic variables x_1 and x_2 are positive.

Definition: A basic feasible solution is said to be **degenerate** if one or more of its basic variables is equal to zero. Thus, although the variable is selected as a basic variable, its value turns out to be zero after solving the system of equations.

For example, consider the basic feasible solution

$$(x_1, x_2, x_3, x_4) = (2, 0, 0, 1),$$

where x_1 and x_2 are chosen as the basic variables. Since one of the basic variables, namely x_2 , is zero, this is a **degenerate basic feasible solution**.

Example 8: Find all the basic solutions of the following system of equations:

$$\begin{aligned}x_1 + 2x_2 + x_3 &= 4, \\2x_1 + x_2 + 5x_3 &= 5.\end{aligned}$$

Also, determine whether the basic solutions are degenerate or non-degenerate.

Solution: The given system of equations is

$$\begin{aligned}x_1 + 2x_2 + x_3 &= 4, \\2x_1 + x_2 + 5x_3 &= 5.\end{aligned}$$

Writing the system in matrix form,

$$AX = b,$$

where

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 5 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Since A is a 2×3 matrix, we choose any two columns of A to form a basis matrix B . There are $\binom{3}{2} = 3$ possible choices.

Case 1: Choosing columns 1 and 2

The basis matrix is $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Since $\det(B) = \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1(1) - 2(2) = -3 \neq 0$, the matrix B is non-singular. The basic variables are $X_B = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Now, $X_B = B^{-1}b$, where $B^{-1} = \frac{1}{-3} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$. Hence,

$$\begin{aligned}X_B &= \frac{1}{-3} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\&= \frac{1}{-3} \begin{bmatrix} -6 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.\end{aligned}$$

Since the remaining variable is $x_3 = 0$, the corresponding basic solution is $\boxed{(2, 1, 0)}$.

Case 2: Choosing columns 1 and 3

The basis matrix is $B = \begin{bmatrix} 1 & 1 \\ 2 & 5 \end{bmatrix}$.

Since

$$\det(B) = \begin{vmatrix} 1 & 1 \\ 2 & 5 \end{vmatrix} = 5 - 2 = 3 \neq 0,$$

B is non-singular. The basic variables are $X_B = \begin{bmatrix} x_1 \\ x_3 \end{bmatrix}$.

Now,

$$B^{-1} = \frac{1}{3} \begin{bmatrix} 5 & -1 \\ -2 & 1 \end{bmatrix}.$$

Therefore,

$$\begin{aligned}X_B &= \frac{1}{3} \begin{bmatrix} 5 & -1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\&= \frac{1}{3} \begin{bmatrix} 15 \\ -3 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}.\end{aligned}$$

Since $x_2 = 0$, the corresponding basic solution is $\boxed{(5, 0, -1)}$.

Case 3: Choosing columns 2 and 3

The basis matrix is $B = \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}$. Since $\det(B) = \begin{vmatrix} 2 & 1 \\ 1 & 5 \end{vmatrix} = 10 - 1 = 9 \neq 0$,

B is non-singular. The basic variables are $X_B = \begin{bmatrix} x_2 \\ x_3 \end{bmatrix}$.

Now,

$$B^{-1} = \frac{1}{9} \begin{bmatrix} 5 & -1 \\ -1 & 2 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} X_B &= \frac{1}{9} \begin{bmatrix} 5 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 15 \\ 6 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{2}{3} \end{bmatrix}. \end{aligned}$$

Since $x_1 = 0$, the corresponding basic solution is $\boxed{\left(0, \frac{5}{3}, \frac{2}{3}\right)}$. **Hence, the three basic**

solutions are $\boxed{(2, 1, 0), \quad (5, 0, -1), \quad \left(0, \frac{5}{3}, \frac{2}{3}\right)}$. Since none of the basic variables is zero in any of the above solutions, all the basic solutions are **non-degenerate**.

Example 9: Show that the feasible solution $x_1 = 1$, $x_2 = 0$, $x_3 = 1$ with $Z = 6$ to the following linear programming problem is not a basic solution:

$$\begin{aligned} \text{Minimize} \quad & Z = 2x_1 + 3x_2 + 4x_3, \\ \text{subject to} \quad & x_1 + x_2 + x_3 = 2, \\ & x_1 - x_2 + x_3 = 2, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Solution: The given feasible solution is $x_1 = 1$, $x_2 = 0$, $x_3 = 1$.

First, we verify that it satisfies the constraints. Substituting the given values into the first equation, $1 + 0 + 1 = 2$, which is satisfied. Similarly, substituting into the second equation, $1 - 0 + 1 = 2$, which is also satisfied. Since

$$x_1, x_2, x_3 \geq 0,$$

the given solution is a feasible solution. The corresponding value of the objective function is

$$Z = 2(1) + 3(0) + 4(1) = 6.$$

Now, let us determine whether this feasible solution is a basic solution.

The coefficient matrix of the system is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}.$$

Since there are two equations, a basic solution must have exactly two basic variables corresponding to a non-singular 2×2 basis matrix. In the given solution, $x_1 = 1$, $x_3 = 1$, are the non-zero variables. Hence, the corresponding basis matrix is formed by the first and third columns of A : $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. Its determinant is $\det(B) = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 1(1) - 1(1) = 0$. Since $\det(B) = 0$, the matrix B is singular and therefore B^{-1} does not exist. Consequently, the columns corresponding to the non-zero variables are linearly dependent and cannot form a basis. Hence, the given feasible solution is *not* a basic solution. Therefore,

The feasible solution $(1, 0, 1)$ with $Z = 6$ is not a basic solution.

Example 10: Determine whether the solution

$$x_1 = 1, \quad x_2 = \frac{1}{2}, \quad x_3 = 0, \quad x_4 = 0, \quad x_5 = 0$$

is a basic solution of the system

$$\begin{aligned} x_1 + 2x_2 + x_3 + x_4 &= 2, \\ x_1 + 2x_2 + x_4 + x_5 &= 2. \end{aligned}$$

Solution: The given system can be written in the matrix form

$$AX = b,$$

where

$$A = \begin{bmatrix} 1 & 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

The given solution is

$$x_1 = 1, \quad x_2 = \frac{1}{2}, \quad x_3 = x_4 = x_5 = 0.$$

First, verify that it satisfies the system.

Substituting into the first equation, $1 + 2\left(\frac{1}{2}\right) + 0 + 0 = 2$. Similarly, substituting into the second equation, $1 + 2\left(\frac{1}{2}\right) + 0 + 0 = 2$. Hence, the given solution satisfies both equations.

Since the system contains $m = 2$ equations, a basic solution must have exactly two basic variables corresponding to a non-singular 2×2 basis matrix.

In the given solution, the non-zero variables are

$$x_1 \quad \text{and} \quad x_2.$$

Therefore, the corresponding basis matrix is formed by the first and second columns of A :

$$B = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}.$$

Now,

$$\det(B) = \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 1(2) - 2(1) = 0.$$

Since

$$\det(B) = 0,$$

the matrix B is singular and hence does not form a basis.

Therefore, although the given solution satisfies the system of equations, it is **not** a basic solution because the columns corresponding to the non-zero variables are linearly dependent.

Hence,

$$\left(1, \frac{1}{2}, 0, 0, 0\right) \text{ is not a basic solution.}$$

Methods for the Solution of a Linear Programming Problem

In general, the following methods are used to solve a Linear Programming Problem (LPP).

1. Graphical (or Geometrical) Method

The graphical method is used when the objective function contains only two decision variables. In this method, the constraints are represented graphically, and the feasible region is obtained. The optimal solution is determined from the feasible region.

Although a problem with three variables can also be represented graphically, it becomes difficult to draw and visualize. Therefore, the graphical method is mainly used for problems involving two variables.

2. Simplex Method

The Simplex Method is the most powerful algebraic method for solving Linear Programming Problems. It can solve problems involving any number of decision variables. The method starts with an initial feasible solution and improves it step by step until the optimal solution is obtained.

The Simplex Method will be discussed in a later chapter.

Graphical Method for Solving a Linear Programming Problem

A Linear Programming Problem having two decision variables can be solved graphically.

There are two techniques for solving an LPP by the graphical method.

1. Corner Point Method.
2. Iso-Profit (or Iso-Cost) Method.

The graphical method is based on the **Fundamental Extreme Point Theorem**, which states that if an optimal solution exists, then it occurs at one of the corner (extreme) points of the feasible region.

Basic Idea

The graphical method consists of the following steps.

1. Draw the graph of each constraint.
2. Determine the region satisfying all the constraints and the non-negative restrictions. This common region is called the **feasible region**.
3. Find all the corner (extreme) points of the feasible region.
4. Evaluate the objective function at each corner point.
5. Select the corner point that gives the maximum (or minimum) value of the objective function.

Procedure for Solving an LPP by the Graphical Method

The following procedure is used.

Step 1. Replace each inequality constraint by an equality.

Step 2. Draw the corresponding straight lines together with the coordinate axes.

To draw the line

$$ax + by = c,$$

find its intercepts by putting

$$x = 0$$

and

$$y = 0.$$

Then join the two intercepts to obtain the straight line.

Step 3. Determine the feasible region.

The feasible region is the common region satisfying all the given constraints together with the non-negative restrictions.

Step 4. Find the optimal solution by one of the following methods.

1. Corner Point Method.
2. Iso-Profit (Iso-Cost) Method.

Working Rule for Finding the Feasible Region

Consider the inequality

$$ax + by \leq c.$$

Draw the corresponding line

$$ax + by = c.$$

This line divides the plane into two regions.

Substitute the test point $(0, 0)$ into the inequality.

- If $(0, 0)$ satisfies the inequality, then the feasible region is the side containing the origin.
- If $(0, 0)$ does not satisfy the inequality, then the feasible region is the side opposite to the origin.

Similarly, for

$$ax + by \geq c,$$

if $(0, 0)$ satisfies the inequality, choose the side containing the origin; otherwise choose the opposite side.

Corner Point Method

The Corner Point Method consists of the following steps.

1. Determine the feasible region.
2. Find all the corner (extreme) points of the feasible region.
3. Compute the value of the objective function at each corner point.
4. Select the corner point giving the maximum (or minimum) value of the objective function.

If two adjacent corner points give the same optimal value, then every point on the line segment joining them is also an optimal solution. In this case, the problem is said to have **infinitely many optimal solutions**.

Iso-Profit (Iso-Cost) Method

An equation of the form

$$Z = ax + by = k,$$

where k is a constant, is called an **iso-profit line** for a maximization problem or an **iso-cost line** for a minimization problem.

To solve the problem,

1. Draw one iso-profit (or iso-cost) line.
2. Draw parallel lines by changing the value of k .
3. For a maximization problem, move the line away from the origin until it touches the feasible region for the last time.
4. For a minimization problem, move the line towards the origin until it first touches the feasible region.
5. The touching point gives the optimal solution.

Note: If the given constraints do not have any common feasible region, then the Linear Programming Problem has **no feasible solution**.

Example 11: Solve the following linear programming problem by the graphical method:

$$\begin{aligned}
 &\text{Minimize } Z = 20x_1 + 10x_2 \\
 &\text{subject to } x_1 + 2x_2 \leq 40, \\
 &\quad 3x_1 + x_2 \geq 30, \\
 &\quad 4x_1 + 3x_2 \geq 60, \\
 &\quad x_1, x_2 \geq 0.
 \end{aligned}$$

Solution: The graphical method consists of the following steps.

Step 1: Convert the inequalities into equations

To draw the boundary lines of the feasible region, replace each inequality by an equality.

$$\begin{aligned}
 x_1 + 2x_2 &= 40, \\
 3x_1 + x_2 &= 30, \\
 4x_1 + 3x_2 &= 60.
 \end{aligned}$$

These equations represent the boundary lines of the given constraints.

Step 2: Find the intercepts of each line

The intercepts required for plotting the lines are given below.

Equation	x_1 -intercept	x_2 -intercept
$x_1 + 2x_2 = 40$	(40, 0)	(0, 20)
$3x_1 + x_2 = 30$	(10, 0)	(0, 30)
$4x_1 + 3x_2 = 60$	(15, 0)	(0, 20)

Using these intercepts, draw the three straight lines on the coordinate plane.

Step 3: Identify the feasible region

Determine the half-plane corresponding to each constraint. Since $x_1 + 2x_2 \leq 40$, the feasible region lies on or below the line $x_1 + 2x_2 = 40$.

Since $3x_1 + x_2 \geq 30$, the feasible region lies on or above the line $3x_1 + x_2 = 30$. Since $4x_1 + 3x_2 \geq 60$, the feasible region lies on or above the line $4x_1 + 3x_2 = 60$. The non-negativity constraints $x_1 \geq 0, x_2 \geq 0$ restrict the feasible region to the first quadrant.

The common region satisfying all the above constraints is the feasible region, i.e., the shaded region $PQRS$ is the feasible region of the problem.

Step 4: Determine the corner points

The corner points of the feasible region are obtained by solving the equations of the intersecting boundary lines.

(i.) The point $P = (40, 0)$ is obtained from the intersection of $x_1 + 2x_2 = 40$ with the x_1 -axis.

(ii.) The point $Q(4, 18)$ is obtained by solving

$$x_1 + 2x_2 = 40,$$

$$3x_1 + x_2 = 30.$$

(iii.) The point $R(6, 12)$ is obtained by solving

$$3x_1 + x_2 = 30,$$

$$4x_1 + 3x_2 = 60.$$

(iv.) The point $S(15, 0)$ is obtained from the intersection of $4x_1 + 3x_2 = 60$ with the x_1 -axis.

Therefore, the vertices of the feasible region are

$$P(40, 0), \quad Q(4, 18), \quad R(6, 12), \quad S(15, 0).$$

Step 5: Evaluate the objective function

The objective function is

$$Z = 20x_1 + 10x_2.$$

Its values at each corner point are given below.

Corner Point	Value of $Z = 20x_1 + 10x_2$
$P(40, 0)$	$20(40) + 10(0) = 800$
$Q(4, 18)$	$20(4) + 10(18) = 260$
$R(6, 12)$	$20(6) + 10(12) = 240$
$S(15, 0)$	$20(15) + 10(0) = 300$

Since this is a minimization problem, we choose the smallest value of the objective function.

$$Z_{\min} = 240,$$

which occurs at

$$R = (6, 12).$$

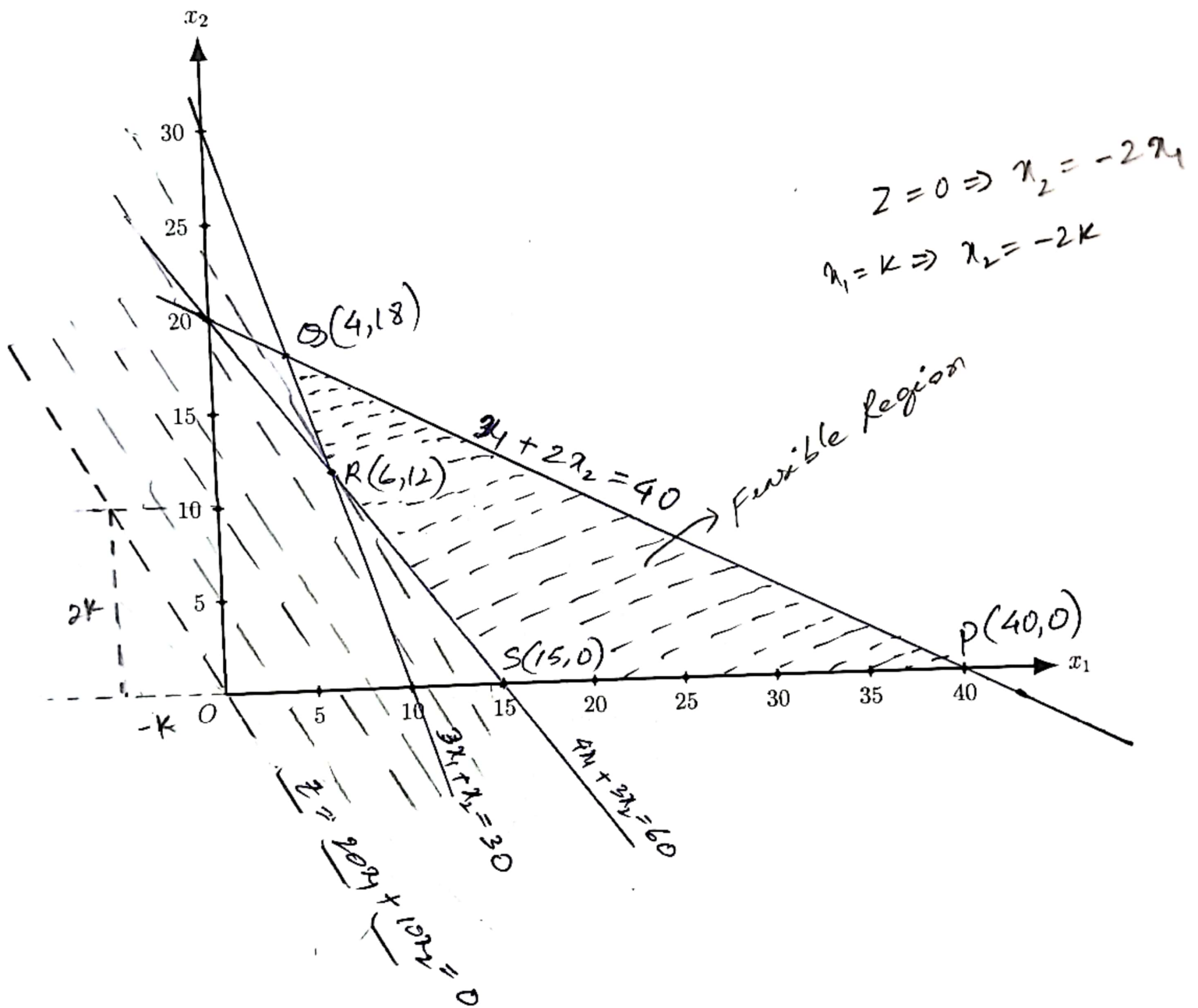
Hence, the optimal solution is

$x_1 = 6,$	$x_2 = 12,$	$Z_{\min} = 240.$
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Solution by the Iso-profit Method

The objective function is $Z = 20x_1 + 10x_2$. To determine the direction of increasing values of Z , first consider $Z = 0$. Then, $20x_1 + 10x_2 = 0$, or equivalently, $2x_1 + x_2 = 0$. This line passes through the origin and has slope -2. Draw this line and then draw parallel lines corresponding to increasing values of Z .

Since the problem is a minimization problem, move the line towards the feasible region from the origin. The first point at which the line touches the feasible region is the point $R(6, 12)$. Therefore, $Z = 20(6) + 10(12) = 240$. Hence, $x_1 = 6, x_2 = 12, Z_{\min} = 240$.



Example 12: Solve the following linear programming problem graphically.

$$\text{Maximize } Z = 8x_1 + 7x_2$$

subject to

$$3x_1 + x_2 \leq 66000,$$

$$x_1 + x_2 \leq 45000,$$

$$x_1 \leq 20000,$$

$$x_2 \leq 40000,$$

$$x_1, x_2 \geq 0.$$

Solution: The graphical method consists of the following steps.

Step 1: Convert the inequalities into equations

To draw the boundary lines of the feasible region, replace each inequality by an equality.

$$3x_1 + x_2 = 66000,$$

$$x_1 + x_2 = 45000,$$

$$x_1 = 20000,$$

$$x_2 = 40000.$$

These equations represent the boundary lines of the given constraints.

Step 2: Find the intercepts of each line

The intercepts required for plotting the lines are given below.

Equation	x_1 -intercept	x_2 -intercept
$3x_1 + x_2 = 66000$	(22000, 0)	(0, 66000)
$x_1 + x_2 = 45000$	(45000, 0)	(0, 45000)
$x_1 = 20000$	(20000, 0)	No intercept
$x_2 = 40000$	No intercept	(0, 40000)

Using these intercepts, draw the four boundary lines on the coordinate plane.

Step 3: Identify the feasible region

Determine the half-plane corresponding to each constraint.

Since $3x_1 + x_2 \leq 66000$, the feasible region lies on or below the line $3x_1 + x_2 = 66000$. Since $x_1 + x_2 \leq 45000$, the feasible region lies on or below the line $x_1 + x_2 = 45000$. Since $x_1 \leq 20000$, the feasible region lies to the left of the line $x_1 = 20000$. Since $x_2 \leq 40000$, the feasible region lies below the line $x_2 = 40000$. The non-negativity constraints $x_1 \geq 0, x_2 \geq 0$ restrict the feasible region to the first quadrant.

The common region satisfying all the above constraints is the feasible region.

Step 4: Determine the corner points

The corner points of the feasible region are obtained by solving the equations of the intersecting boundary lines.

(i.) The point $O = (0, 0)$ is obtained from the intersection of the coordinate axes.

(ii.) The point $A = (0, 40000)$ is obtained from the intersection of $x_2 = 40000$ with the x_2 -axis.

(iii.) The point $B = (5000, 40000)$ is obtained by solving

$$\begin{aligned}x_1 + x_2 &= 45000, \\x_2 &= 40000.\end{aligned}$$

(iv.) The point $P = (10500, 34500)$ is obtained by solving

$$\begin{aligned}3x_1 + x_2 &= 66000, \\x_1 + x_2 &= 45000.\end{aligned}$$

(v.) The point $C = (20000, 6000)$ is obtained by solving

$$\begin{aligned}3x_1 + x_2 &= 66000, \\x_1 &= 20000.\end{aligned}$$

(vi.) The point $D = (20000, 0)$ is obtained from the intersection of $x_1 = 20000$ with the x_1 -axis. Therefore, the vertices of the feasible region are

$$O(0, 0), A(0, 40000), B(5000, 40000), P(10500, 34500), C(20000, 6000), D(20000, 0).$$

Step 5: Evaluate the objective function

The objective function is $Z = 8x_1 + 7x_2$.

Its values at each corner point are given below.

Corner Point	Value of $Z = 8x_1 + 7x_2$
$O(0, 0)$	0
$A(0, 40000)$	280000
$B(5000, 40000)$	320000
$P(10500, 34500)$	325500
$C(20000, 6000)$	202000
$D(20000, 0)$	160000

Since this is a maximization problem, we choose the largest value of the objective function. $Z_{\max} = 325500$, which occurs at $P = (10500, 34500)$. Hence, the optimal solution is $x_1 = 10500, x_2 = 34500, Z_{\max} = 325500$.

Solution by the Iso-profit Method

The objective function is $Z = 8x_1 + 7x_2$. To determine the direction of increasing values of Z , first consider $Z = 0$. Then, $8x_1 + 7x_2 = 0$, or equivalently, $x_2 = -\frac{8}{7}x_1$.

This line passes through the origin and has slope $-\frac{8}{7}$.

Draw this line and then draw parallel lines corresponding to increasing values of Z .

Since the problem is a maximization problem, move the line parallel to itself away from the origin. The last point at which the line touches the feasible region is the point $P(10500, 34500)$.

Therefore,

$$Z = 8(10500) + 7(34500) = 325500.$$

Hence,

$$x_1 = 10500, \quad x_2 = 34500, \quad Z_{\max} = 325500.$$

